E103 Stars and the Universe

E1.3.1 Degrees, arcminutes and arcseconds

One degree contains 60 arcminutes and one arcminute contains 60 arcseconds, so one degree equals 3600 arcseconds. These are angular units, distinct from minutes and seconds of time. Parallax formulae often require arcseconds, whereas small-angle geometrical relations often require radians; check and convert before substitution.

$$1^\circ = 60' = 3600''$$

$$1' = 60''$$

Angular units: degree, arcminute and arcsecond. These are angles, not units of time.

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E1.3.2 · E1.3.4 · E1.3.3 Annual parallax and stellar distances

E1.3.2 · E1.3.4 The geometrical meaning of a parsec

A parsec is the distance corresponding to an annual parallax of one arcsecond. The geometrical baseline is 1 AU, giving approximately 206265 AU or 3.26 light-years by the small-angle relation. An arcsecond measures angle, not time; a parsec measures distance despite the historical reference to seconds in its name.

$$1\ \text{pc} \approx 206\,265\ \text{AU}$$

$$1\ \text{pc} \approx 3.09 \times 10^{16}\ \text{m} \approx 3.26\ \text{ly}$$

Small-angle parallax geometry: one radian is approximately 206 265 arcseconds.

Parsec: distance corresponding to annual parallax of one arcsecond. Rounded conversions.

E1.3.3 Distance from annual parallax

As Earth orbits the Sun, a nearby star appears to shift relative to distant stars. Annual parallax is half the maximum total angular shift measured six months apart. Distance in parsecs is the reciprocal of parallax in arcseconds. More distant stars have smaller parallaxes; do not substitute the full angular shift.

$$\frac{d}{\text{pc}} = \frac{1}{p/\text{arcsec}}$$

d: distance in parsecs; p: annual parallax angle in arcseconds, half the total angular shift over six months. This numerical relation requires these units.

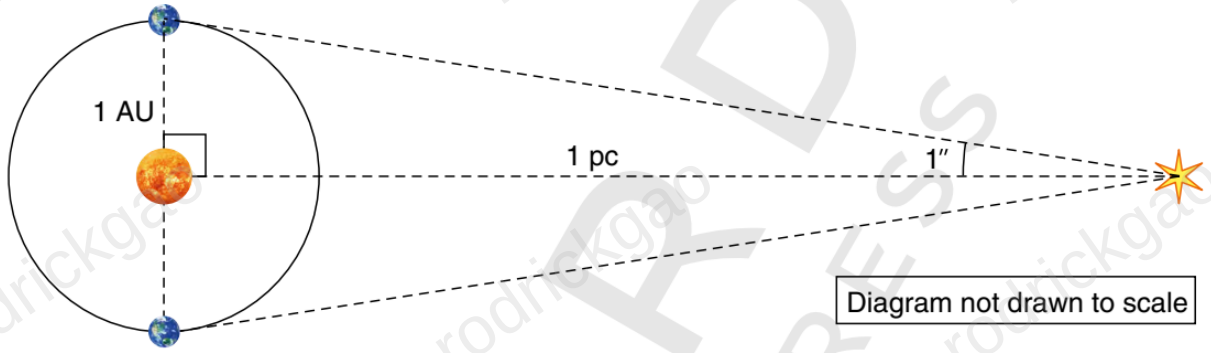


Fig 3.1i *Definition of one parsec.*

The one-AU baseline subtends the annual parallax angle. The full apparent shift over six months is twice that angle.

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E1.3.5 · E1.3.6 Luminosity and apparent brightness

Luminosity is a star's total energy emitted per second in all directions. Apparent brightness is power received per unit area at the observer. For isotropic emission with negligible absorption, power spreads over a sphere, so doubling distance quarters apparent brightness while leaving luminosity unchanged.

$$L = \frac{E_{\text{emitted}}}{\Delta t}$$

$$b = \frac{L}{4\pi d^2}$$

Luminosity L (W): total radiated energy (J) per time (s), emitted in all directions.

Apparent brightness b (W m⁻²); luminosity L (W); distance d (m). Isotropic emission and negligible absorption.

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E1.3.7 · E1.3.8 Apparent magnitude

Apparent magnitude describes observed brightness, with smaller numbers indicating brighter objects. It depends on intrinsic luminosity, distance and possibly absorption along the path. A brighter-looking star may simply be closer rather than more luminous. Apparent magnitude is not absolute magnitude.

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E1.3.9 · E1.3.10 Absolute magnitude

Absolute magnitude is the apparent magnitude an object would have at 10 pc, allowing intrinsic brightness to be compared at a common distance. Smaller values indicate greater intrinsic brightness, irrespective of actual distance from Earth. Compare the same wavelength band or a consistent bolometric magnitude definition.

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E1.3.11 · E1.3.12 · E1.3.13 · E1.3.14 Black-body radiation and stars

E1.3.11 · E1.3.12 Black-body spectra and stellar colour

For an approximate black body, increasing temperature shifts the spectral peak to shorter wavelengths and increases total emission. Hotter stars therefore appear bluer and cooler stars redder. Colour reflects the overall spectrum seen by the eye, not emission at one wavelength only. Use absolute temperature in the relevant relations.

$$T \uparrow \Longrightarrow \lambda_{\max} \downarrow$$

Blackbody spectrum: increasing absolute temperature T shifts the wavelength of peak emission $\lambda_{\max}$ towards shorter wavelengths.

E1.3.13 · E1.3.14 Black-body radiation power

Total emitted power per unit surface area of a black body is proportional to kelvin temperature to the fourth power. Multiply by a spherical star's entire surface area to obtain luminosity. Doubling radius quadruples area; doubling temperature increases power per area sixteenfold. Net exchange also subtracts absorbed power.

$$J = \sigma T^4$$

$$L = 4\pi R^2 \sigma T^4$$

J: total radiant power per surface area (W m^{-2}); T: absolute temperature (K); σ = 5.67 × 10^{-8} W m^{-2} K^{-4}. Blackbody emission, not net exchange with surroundings.

Spherical blackbody: L: luminosity (W); R: stellar radius (m); T: surface temperature (K); σ = 5.67 × 10^{-8} W m^{-2} K^{-4}.

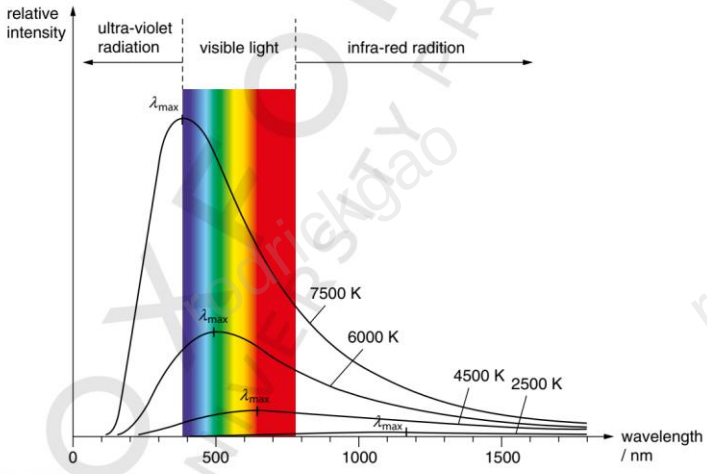


Fig 3.3d *Blackbody radiation curves at different surface temperatures. A visible spectrum is added to indicate the wavelengths of visible light.*

A hotter blackbody has greater total emission and a shorter peak wavelength. The peak position is not itself the perceived colour of the complete spectrum.

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E1.3.15 Stellar absorption spectra

Gas in a star's outer layers absorbs particular wavelengths and re-emits in all directions, reducing their intensity along the observer's line of sight and producing dark lines. Each element has characteristic line patterns, allowing outer-atmosphere elements to be identified. One line alone does not determine the whole star's elemental proportions.

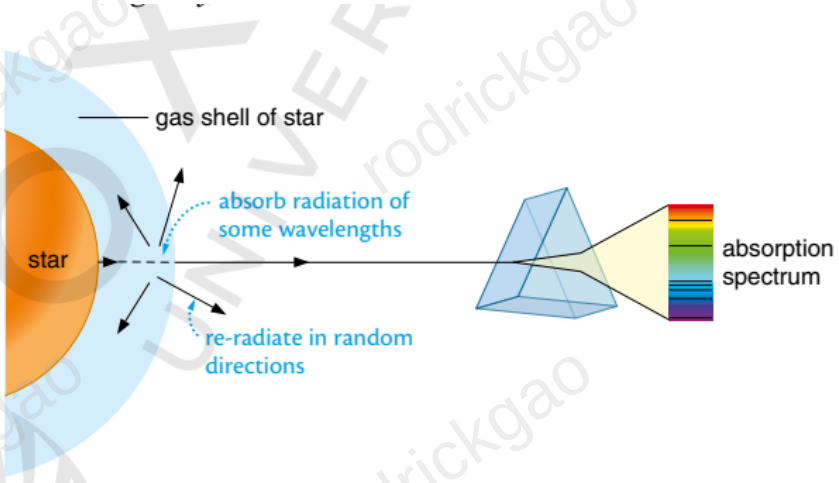


Fig 3.3j *Formation of absorption spectrum.*

Outer gas absorbs particular wavelengths, producing dark lines in the stellar spectrum.

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E1.3.16 · E1.3.17 Stellar properties and the H–R diagram

E1.3.16 Spectral classes and the H–R diagram

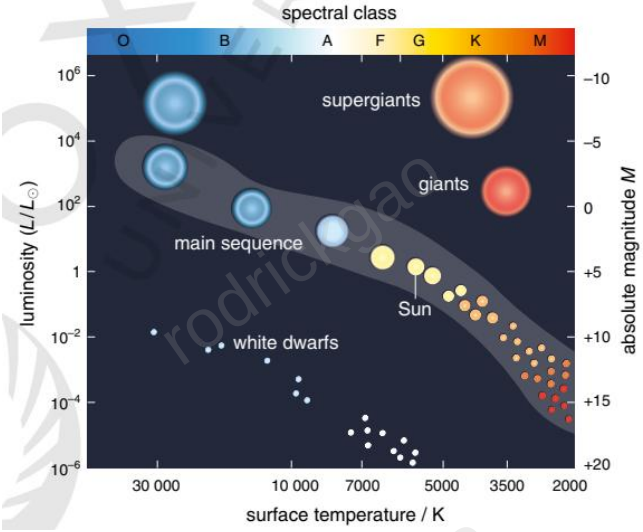
Spectral types run O, B, A, F, G, K, M from hottest to coolest. In a typical H–R diagram, temperature decreases to the right and luminosity increases upwards; on an absolute-magnitude axis, smaller values mean brighter stars. The main sequence slopes from upper left to lower right, giants lie higher, and white dwarfs lie lower left: hot but faint and small. Check axis quantities and directions before identifying populations.

E1.3.17 Comparing stellar radius, temperature and luminosity

For spherical black-body stars, luminosity ratio equals radius ratio squared times absolute-temperature ratio to the fourth power. Comparing with the Sun cancels shared constants. A large cool star can outshine a small hot one. Colour alone cannot determine luminosity or radius; combine temperature with another relevant quantity.

$$\frac{L}{L_\odot} = \left(\frac{R}{R_\odot}\right)^2 \left(\frac{T}{T_\odot}\right)^4$$

Compare spherical blackbody stars with the Sun (⊙). L: luminosity; R: radius; T: kelvin temperature. Corresponding quantities use identical units.



An H–R diagram: temperature decreases to the right and luminosity increases upwards.

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E1.3.18 Doppler shifts of spectral lines

Recession along the line of sight shifts spectral lines to longer wavelengths, called redshift; approach shifts them shorter, called blueshift. At radial speeds much smaller than light speed, fractional wavelength change estimates radial velocity. It gives only the line-of-sight component, not total space speed. Compare the same line with its rest wavelength.

$$\frac{\Delta\lambda}{\lambda_0} = \frac{\lambda - \lambda_0}{\lambda_0} \approx \frac{v_r}{c}$$

Non-relativistic radial Doppler shift: radial speed is much smaller than c. λ_0: rest wavelength; λ: observed wavelength; $v_r > 0$: receding (redshift); $v_r < 0$: approaching.

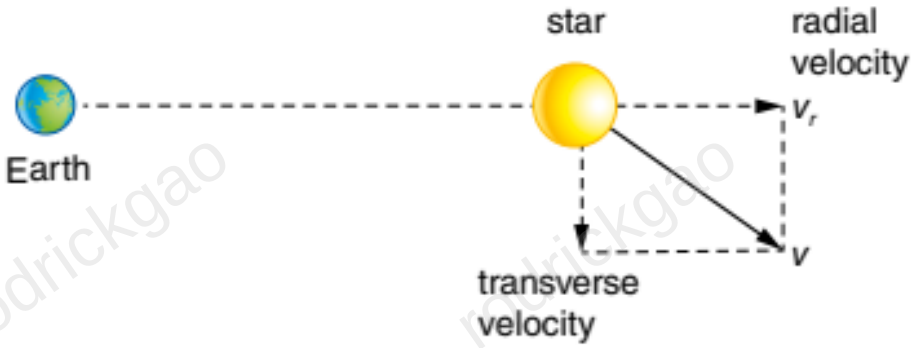


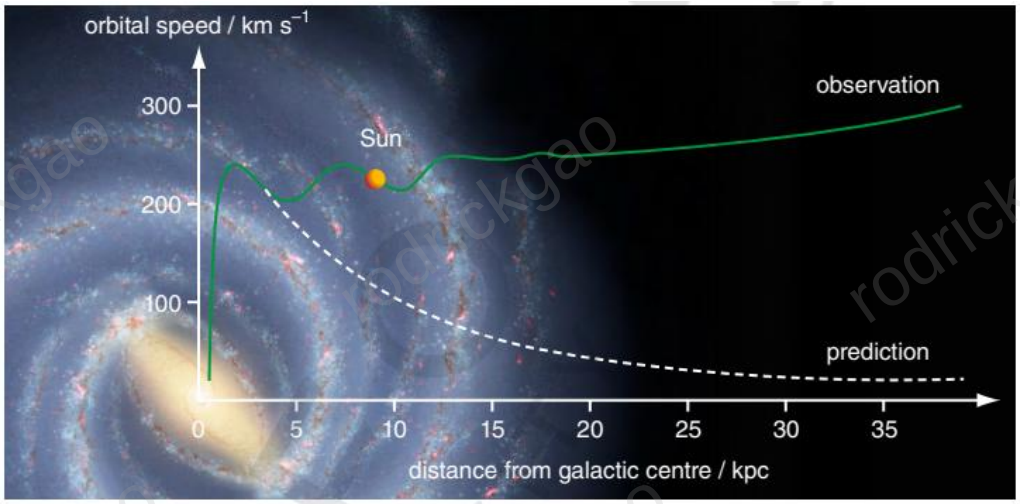
Fig 3.4e *Transverse and radial velocities of a star.*

Doppler measurements determine line-of-sight radial velocity, not transverse velocity.

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E1.3.19 Galaxy rotation curves and dark matter

If a galaxy's mass were predominantly visible matter concentrated near its centre, outer stars would be expected to orbit more slowly. The textbook curves instead show similar or increasing outer speeds, suggesting additional unseen mass supplying gravity. These are orbital speeds around a galactic centre, not Hubble recession speeds.



Observed outer orbital speeds differ from predictions using only visible mass.

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E1.3.20 Hubble's relation

On cosmological scales, more distant galaxies have greater recession speeds on average, supporting an expanding Universe. This is not orbital speed around a galactic centre. Nearby galaxies need not all recede, because local gravitational motions can produce departures from the large-scale relation.

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